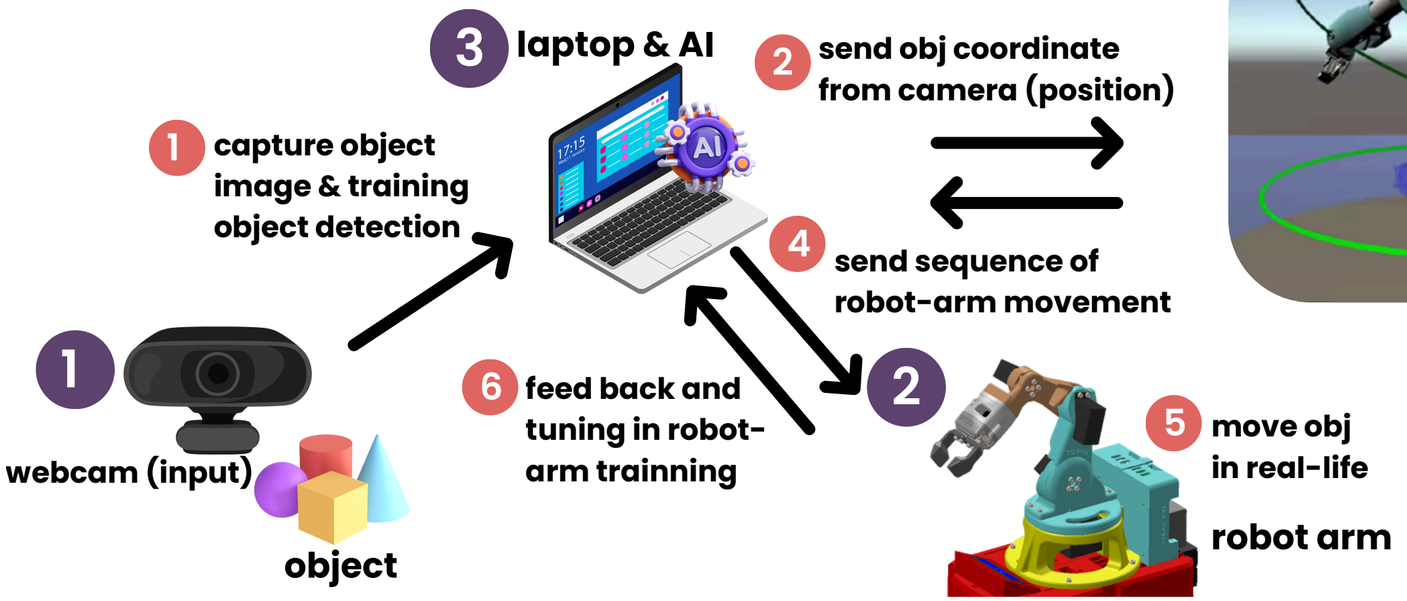
Week09

Reinforcement learning for Control RobotArm

RL for Digital Twin

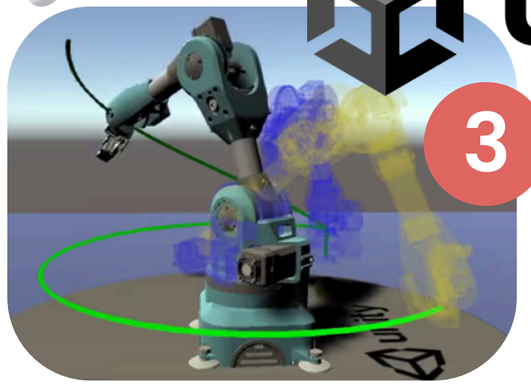
Are you ready?

Content of this week



WORKSHOP 4

Unity

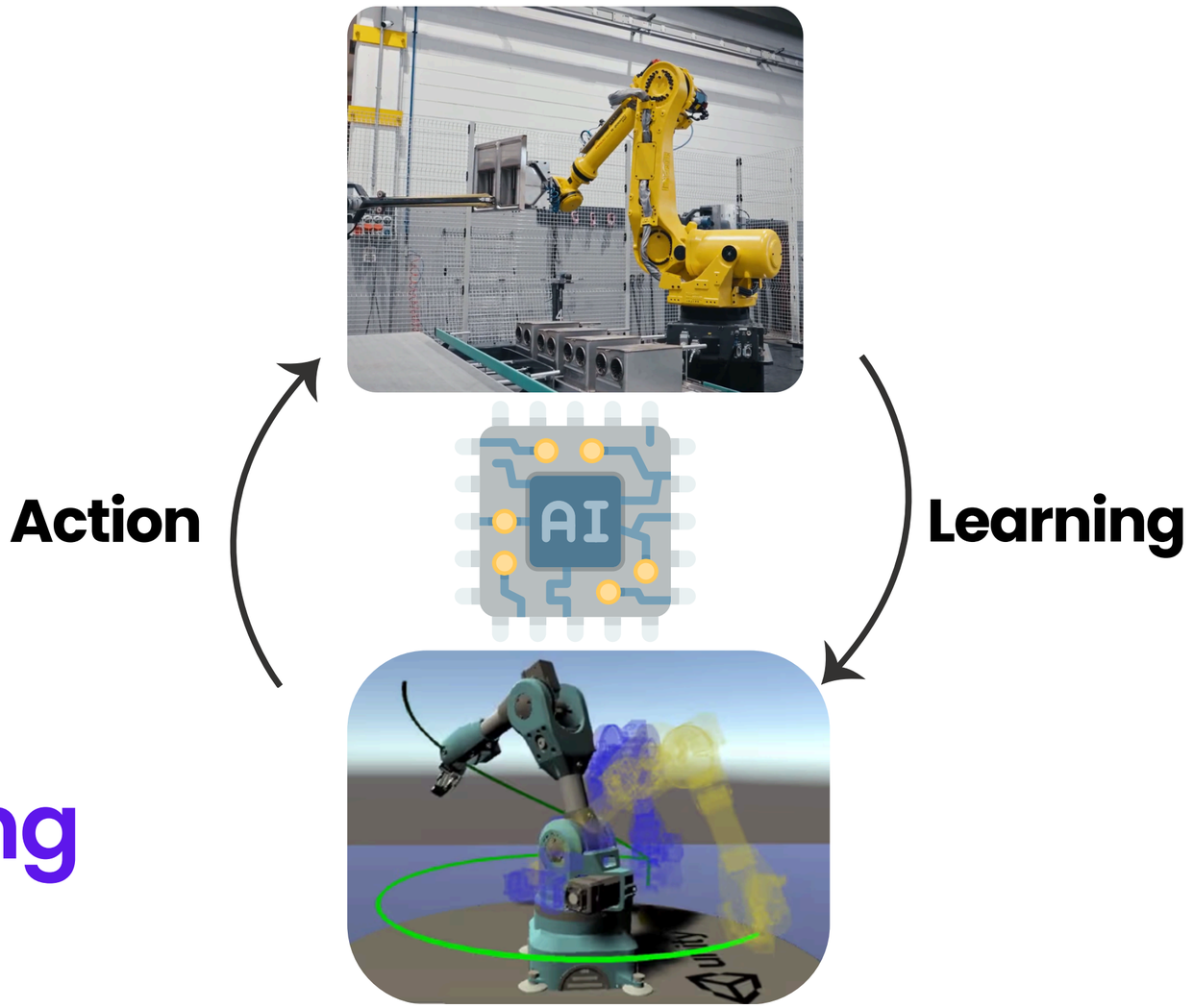


3 training move obj in unity

RECAP

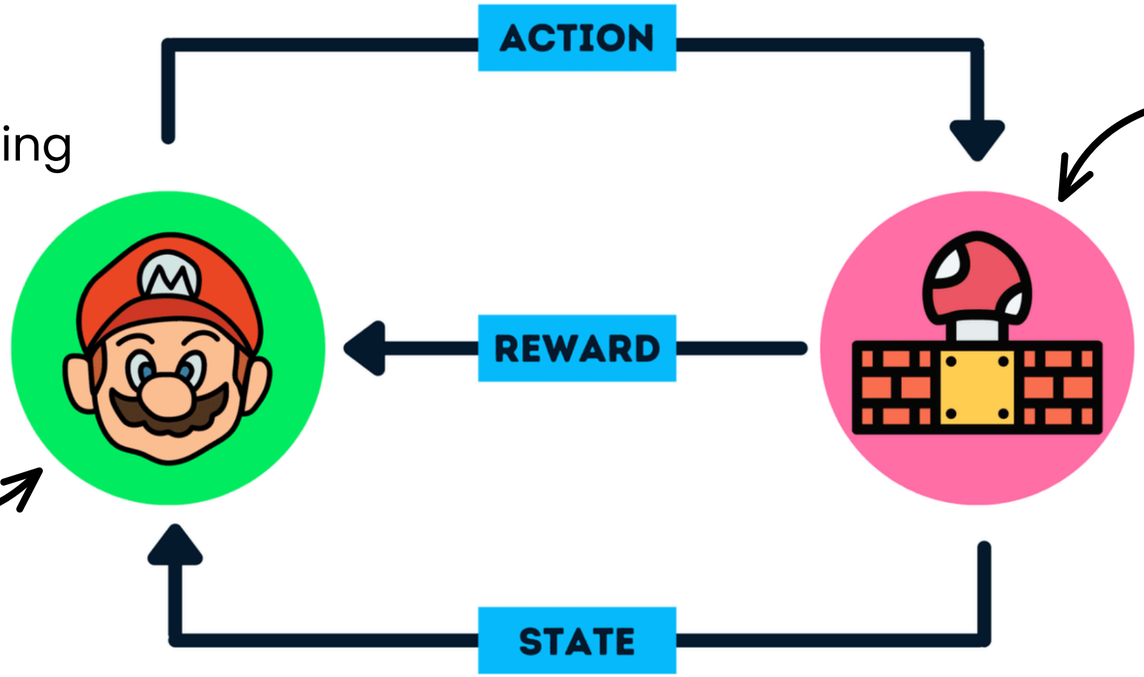
What tool's using for training automated Robot Arm

Reinforcement learning



recap-component of RL

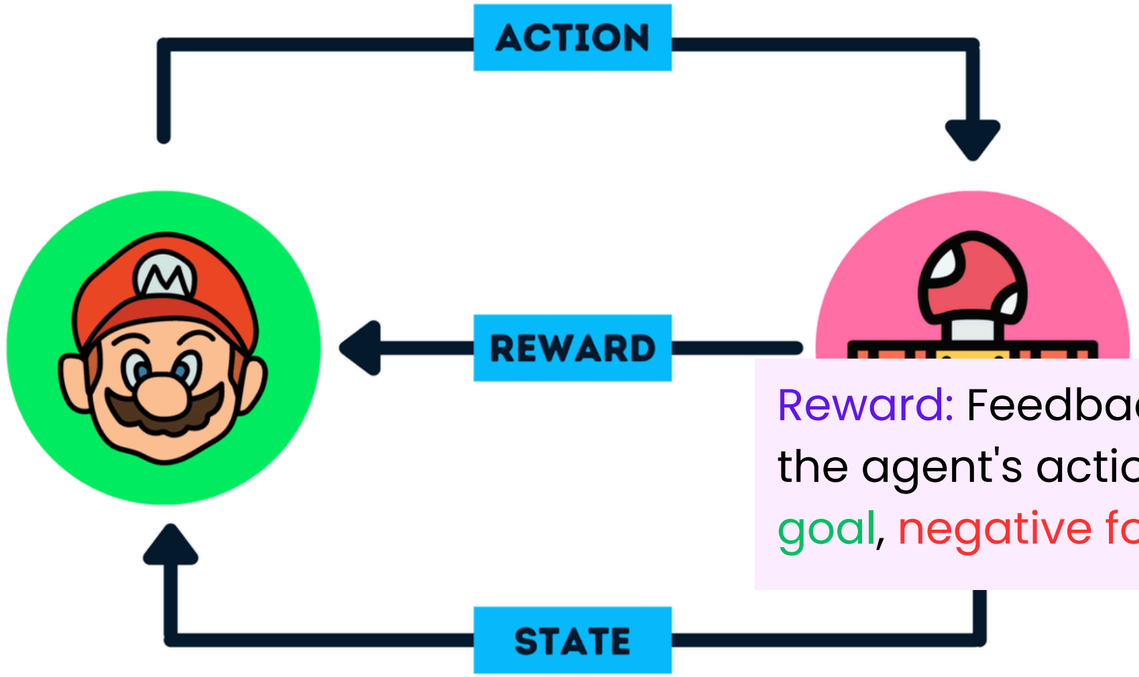
Agent: The AI or decision-maker learning to perform a task (e.g., a robot, a self-driving car).



Environment: The space or system the agent interacts with.

recap-component of RL

Action: The moves or decisions the agent can take.



Reward: Feedback from the environment evaluating the agent's action (e.g., positive for moving closer to a goal, negative for hitting an obstacle).

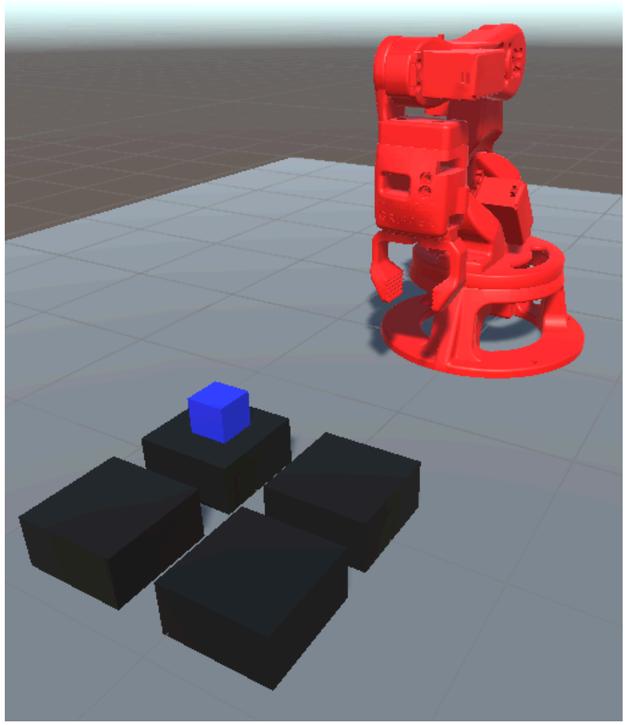
State: The current condition or situation of the environment perceived by the agent.

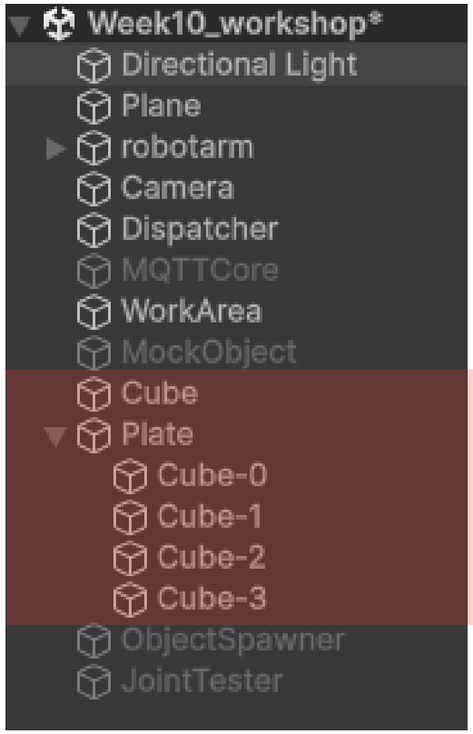
WORKSHOP

Prepare environment
Training RebotArm
unity-ml.agent

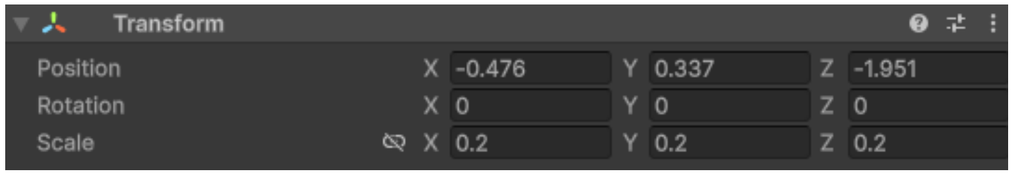
https://github.com/idektep/DGT5/tree/week9

Step 1: prepare environment



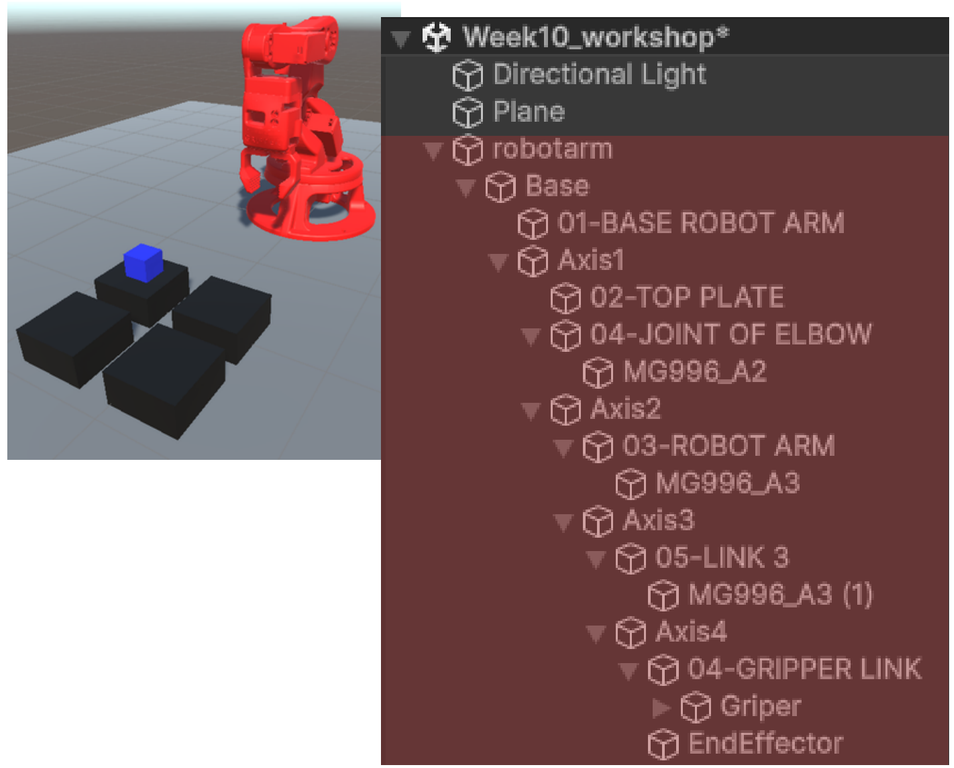


STEP 1 — สร้าง object
เมนู GameObject → 3D Object → Cube → ชื่อ Cube
Transform → Scale = (1, 1, 1)



STEP 2 — สร้าง Plate (Cube-0 to Cube-4)
Create Empty → ชื่อ Plate
เมนู GameObject → 3D Object → Cube → ชื่อ Cube-0
Transform → Scale = (0.5, 0.5, 0.5)

Step 2: prepare joint RobotArm

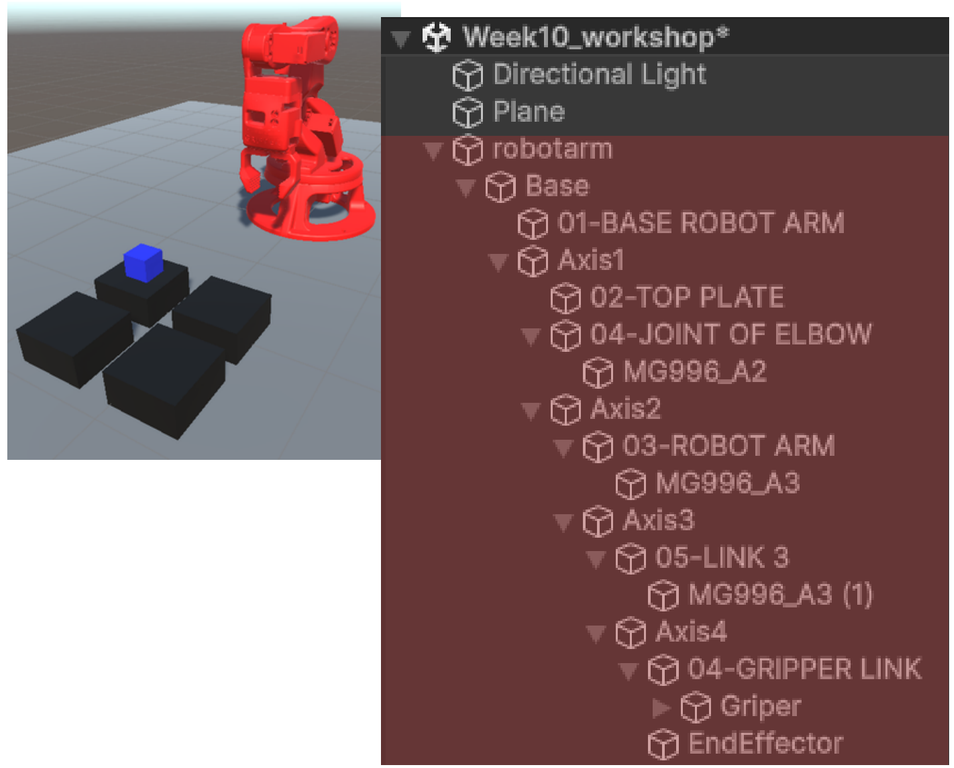


Check Transform of Axis1-Axis4

- **Axis1 rotate in ?**
- **Axis2 rotate in ?**
- **Axis3 rotate in ?**
- **Axis4 rotate in ?**

Let's Testing and Recheck
5 mins

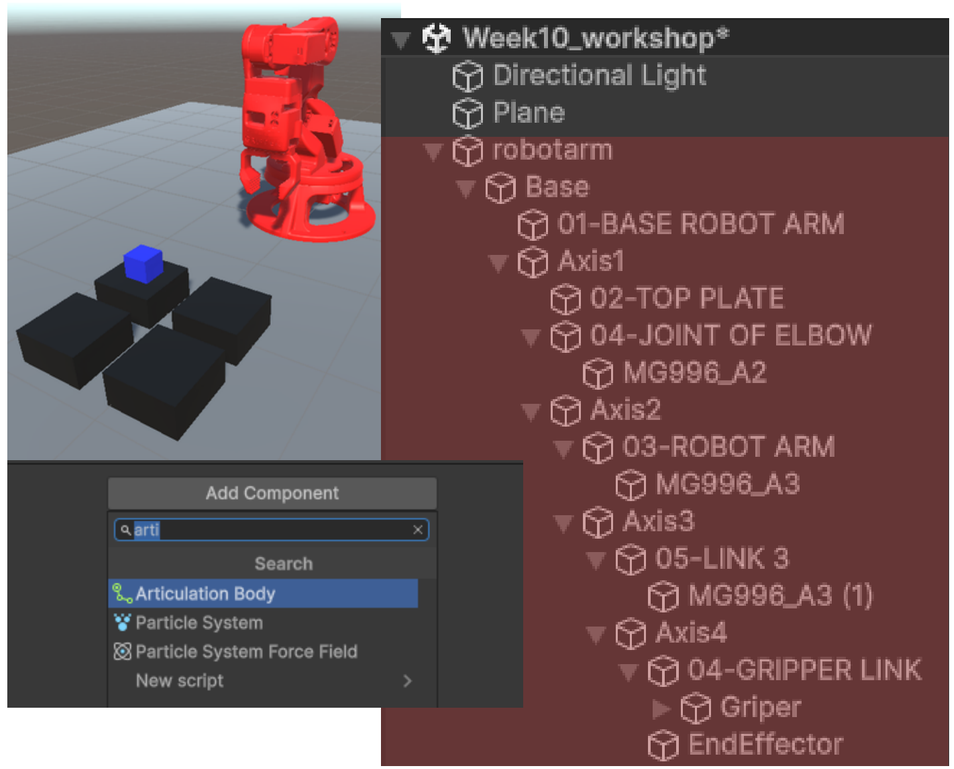
Step 2: prepare joint RobotArm



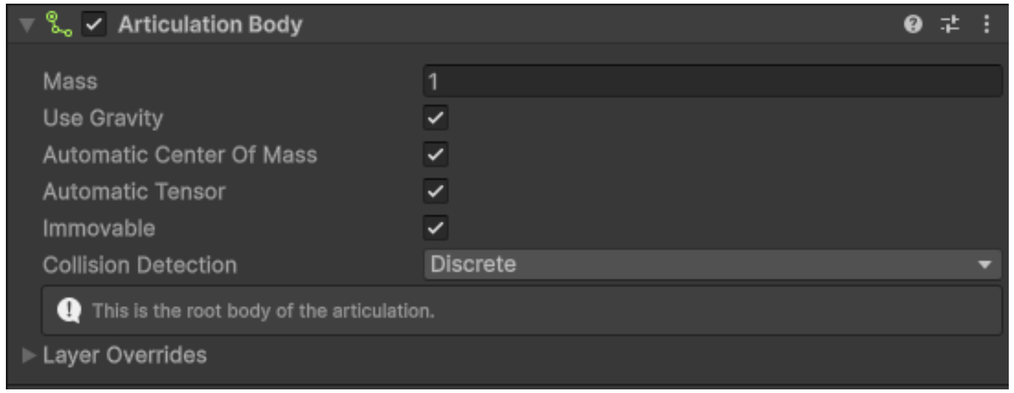
Check Transform of Axis1-Axis4

- **Axis1 rotate in Y**
- **Axis2 rotate in X**
- **Axis3 rotate in X**
- **Axis4 rotate in X**

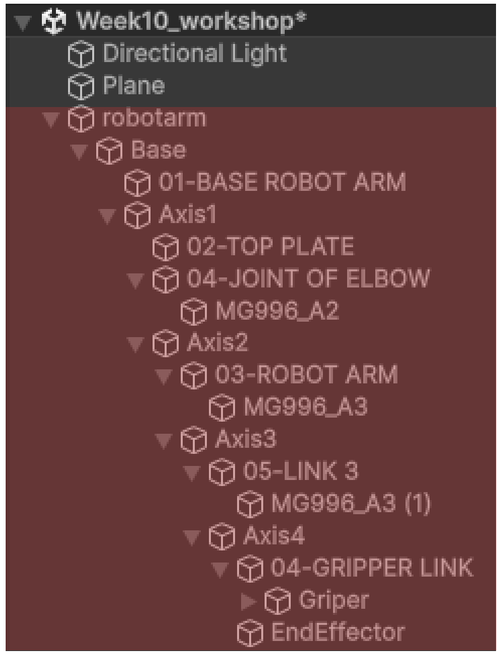
Step 2: prepare joint RobotArm

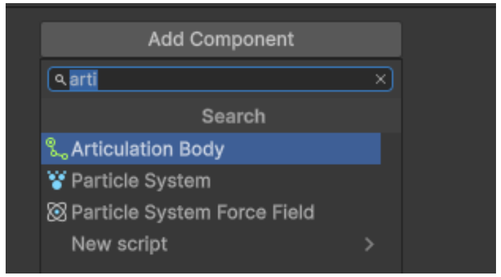


STEP — เลือก Base
Add Component → Articulation Body →
Check Immovable



Step 2: prepare joint RobotArm

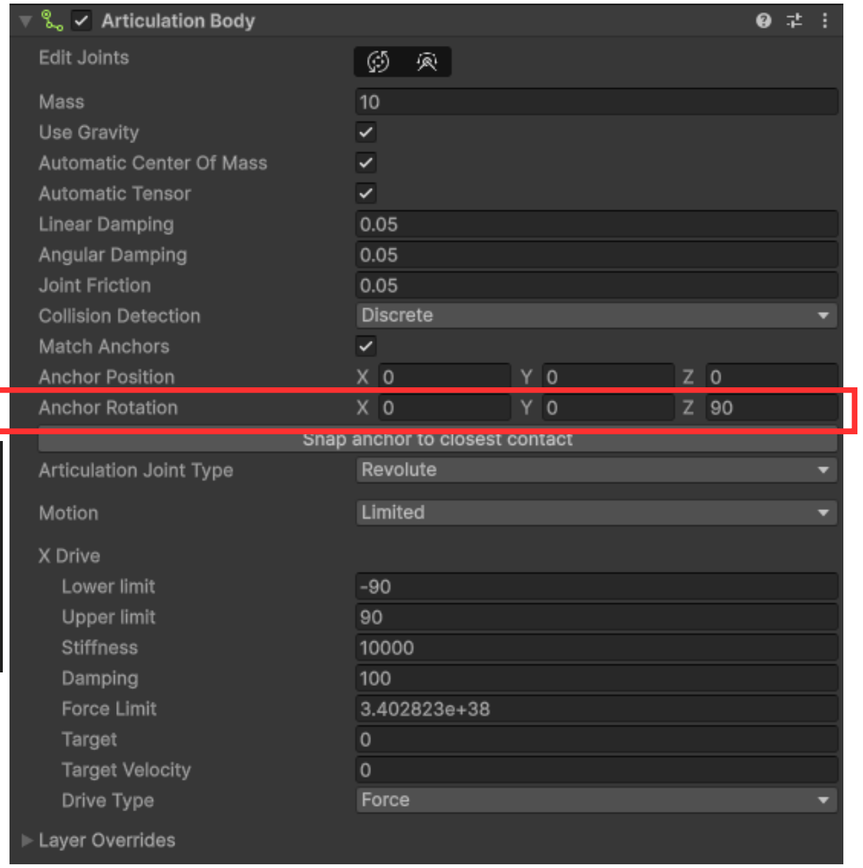




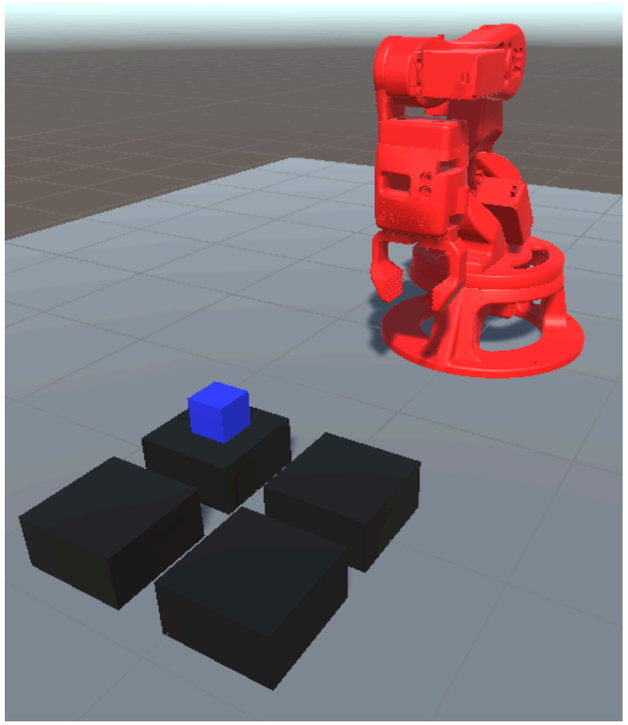
STEP — เลือก Axis1
Add Component → Articulation Body
→ Config by follow this setting

STEP — Axis2, Axis3, Axis4: repeat this step of Axis1 on each

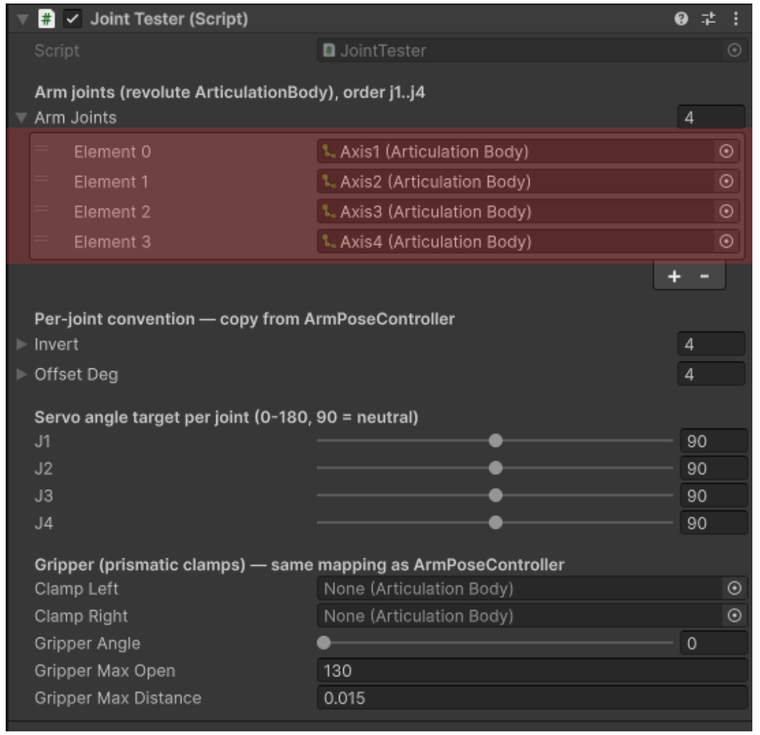
Joint	Joint Type	หมุนรอบ	Anchor Rotation (เริ่มต้น)	Lower / Upper (ใน X Drive)
Axis1	Revolute	Y	(0, 0, 90)	−90 / 90
Axis2	Revolute	X	(0, 0, 0)	−90 / 90
Axis3	Revolute	X	(0, 0, 0)	−90 / 90
Axis4	Revolute	X	(0, 0, 0)	−90 / 90



Step 2: prepare joint RobotArm

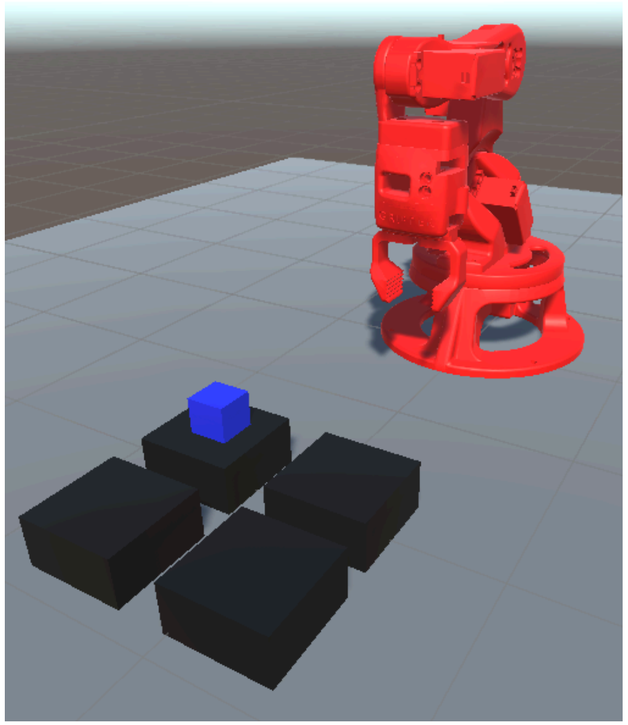


STEP 1 — Hierarchy → Create Empty → ตั้งชื่อ JointTester → Add Component → JointTester

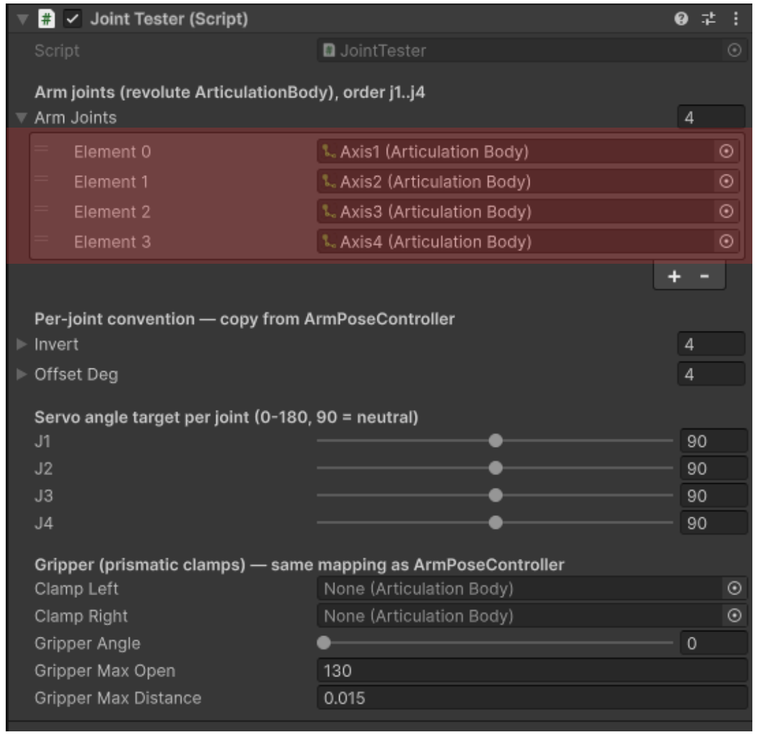


STEP — กด play → เลือก JointTester และปรับ J1-J4 สำหรับเช็คว่า robot arm ขยับถูกต้อง

Step 2: prepare joint RobotArm

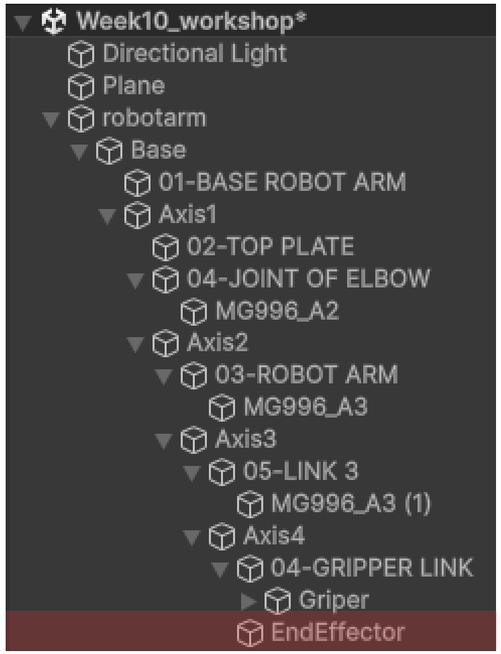


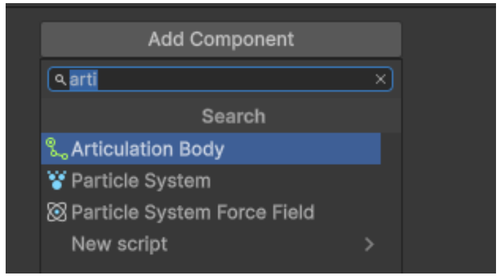
STEP 1 — Hierarchy → Create Empty → ตั้งชื่อ JointTester → Add Component → JointTester



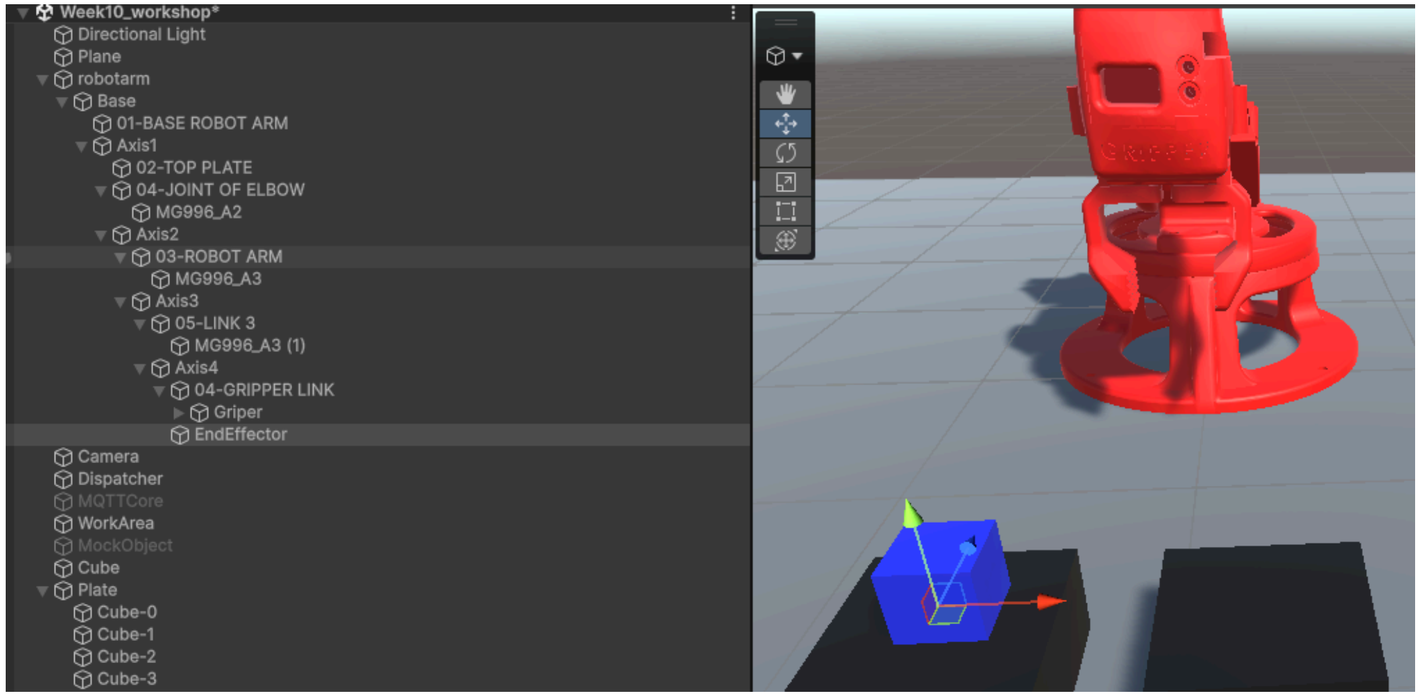
STEP — กด play → เลือก JointTester และปรับ J1-J4 สำหรับเช็คว่า robot arm ขยับถูกต้อง

Step 3: Add markers

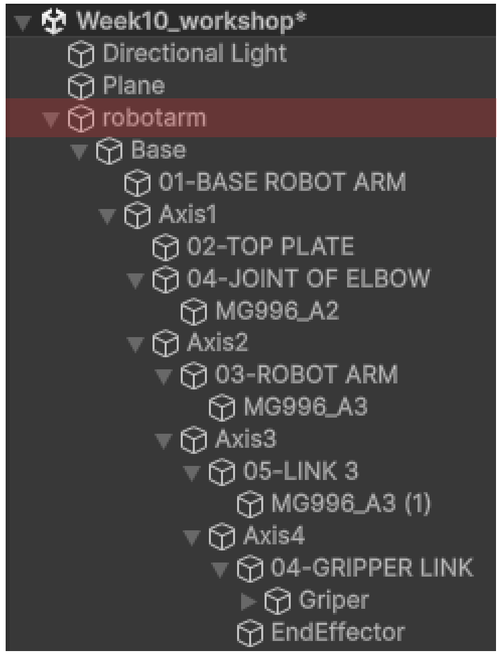




STEP — เลือก Axis4 → Create Empty → ตั้งชื่อ EndEffector → ขยับ box ให้อยู่ควบคู่กับ Object.Cube และวางไว้ที่ Plate slot0 (Cube-0)



Step 4: Setup for training robotarm



STEP — เลือก robotarm → Add Component→

- ArmAgent
- Behavior Parameters
- Decision Requester

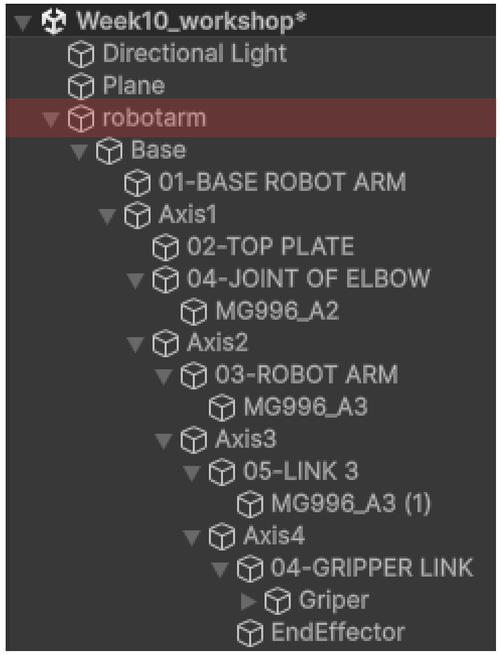
3.2 — Behavior Parameters

Field	Value
Behavior Name	`ArmReach`
Vector Observation → Space Size	11
Actions → Continuous Actions	4
Model	None (empty)
Behavior Type	Default

3.3 — Decision Requester

- **Decision Period** = 5 (leave "Take Actions Between Decisions" checked)

Step 4: Setup robotarm

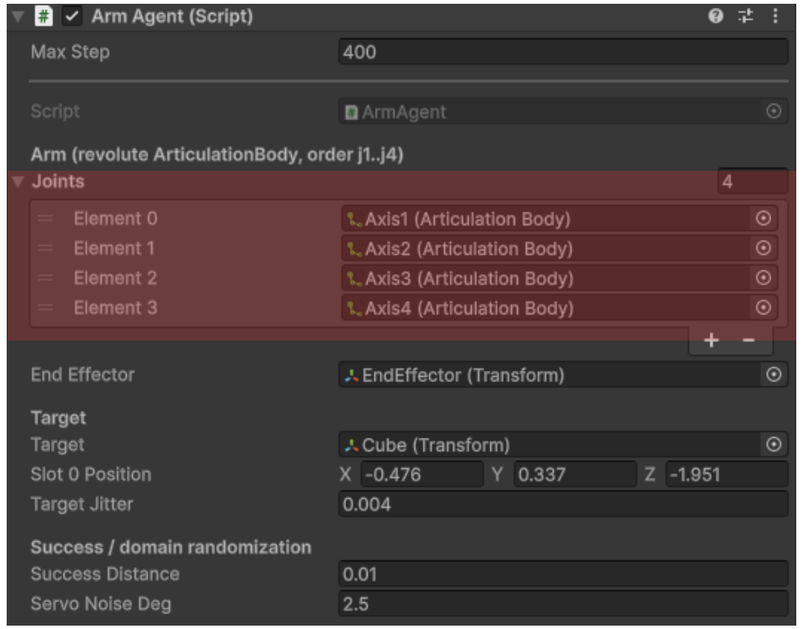


STEP — เลือก robotarm → Add Component→

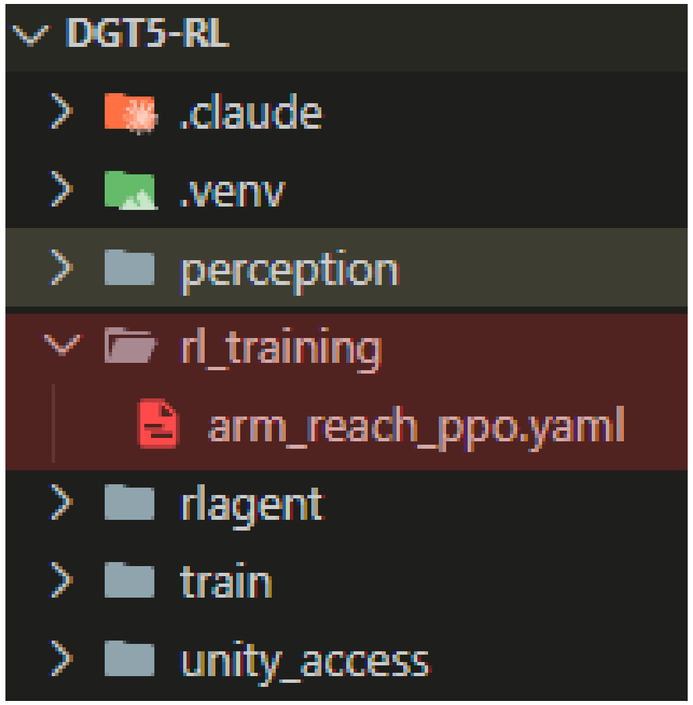
- ArmAgent
- Behavior Parameters
- Decision Requester

3.4 — ArmAgent (wire it)

Field	Value
joints	Size 4 → Axis1, Axis2, Axis3, Axis4 (their ArticulationBodies)
endEffector	EndEffector
target	MockObject
slot0Position	Cube-0's X, Y, Z (optionally +0.01 on Y so the tip stops at object height)
targetJitter	0.004
successDistance	0.01
servoNoiseDeg	2.5
Max Step	**400**



Step 5: write training config

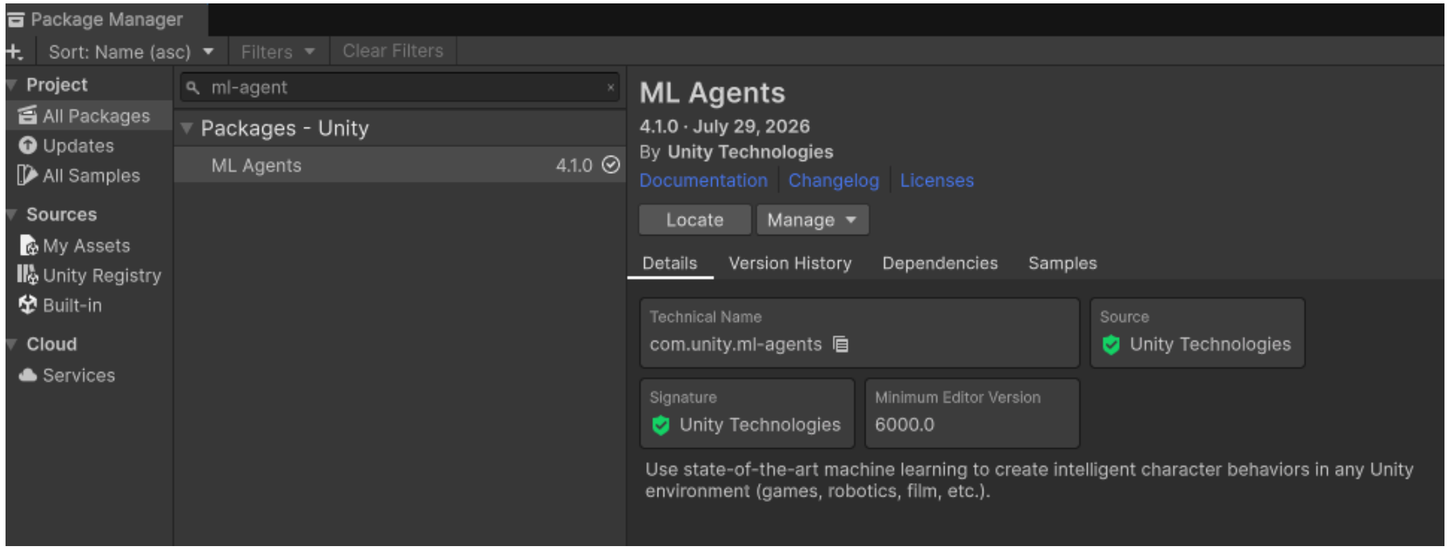


STEP — สร้าง folder “rl_training” → สร้างไฟล์ arm_reach_ppo.yaml

```yaml
# PPO config
# component "Behavior Name" in the Behavior Parameters in Unity ("ArmReach").
# Run:  mlagents-learn rl_training/arm_reach_ppo.yaml --run-id=arm_reach_1
# then press Play in Unity.
behaviors:
  ArmReach:
    trainer_type: ppo
    hyperparameters:
      batch_size: 1024
      buffer_size: 10240
      learning_rate: 3.0e-4
      beta: 5.0e-3
      epsilon: 0.2
      lambd: 0.95
      num_epoch: 3
      learning_rate_schedule: linear
    network_settings:
      normalize: true
      hidden_units: 256
      num_layers: 2
    reward_signals:
      extrinsic:
        gamma: 0.99
        strength: 1.0
    max_steps: 2000000
    time_horizon: 64
    summary_freq: 10000
```

Step 6: prepare for training

STEP — สร้าง folder “rl_training” → สร้างไฟล์ arm_reach_ppo.yaml



STEP — To add the ML-Agents package to a Unity project:
Open the Package Manager, navigate to Window → Package Manager.
Click + and select Add package by name...
Enter **com.unity.ml-agents** *Click Add to add the package to your project.

Step 7: prepare RL environment

STEP — download python version 3.10.10

www.python.org/downloads/release/python-31010

Install : add Path , Install Python 3.10 for all user

STEP — In VSCODE → new terminal → deactivate .venv

ย้อนกลับ(cd ..) มาจนถึง path “../dgt5-rl”

STEP — create rl-venv

: py --list

: py -3.10 -m venv rlagent

→ activate rlagent

Install package : pip install mlagents

Step 7: prepare RL environment

VS Code

STEP — In VSCODE Run command to start training RL model

cd C:\...\dgt5-rl

mlagents-learn rl_training/arm_reach_ppo.yaml --run-id=arm_reach_1

STEP — Wait until it prints the Unity logo and:
[INFO] Listening on port 5004. Start training by pressing the Play button in the Unity Editor.

Unity

STEP — Press ▶ Play in Unity. Training begins

COMPLETE.

Step 7: prepare RL environment

The error
ModuleNotFoundError: No module named 'pkg_resources'
Fix In your activated env (rlagent):
pip install "setuptools<81"

VS Code

- **Watch progress**

In the terminal, every 10,000 steps you'll see a line like:
: Step: 10000. Time Elapsed: 30s. Mean Reward: -0.85. Std of Reward: 0.4.

Run again / continue
New run: change the id — --run-id=arm_reach_2 (reusing an id errors).
Continue the same run: add --resume.
Overwrite the same run: add --force.

COMPLETE.

Thank You For Attention

See You Next Time